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End Semester Examination 2024

Name of the Course: B. Tech

Semester: VI

Name of the Paper – Power System Protection

Paper Code: TEE 603

Time: 3Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)						CO1	
(a)	What is a protective relay? Explain its function in an electrical system.							
(b)	Why protection system is provided in a power system? What is a Unit Protection?							
(c)	Why Instrument transformers are required in protective relaying scheme?							
Q2	(20 marks)						CO2	
(a)	Why is the reset to pick-up ratio low in dc attracted armature type relay as compared to ac relay? Explain with reasons why an unpleasant noise (relay chattering) sometimes occurs in an ac attracted armature type relay. Discuss the remedial measures adopted to minimize this unpleasant noise.							
(b)	The current setting is set at 75%, TMS = 0.4; C.T. Ratio = 400/5, fault current is 6000 A. Determine the operating time of the relay. At TMS = 1 the operating time at various PSMs are shown in the table below :							
	PSM	2	4	5	8	10		20
	Operating time in secs.	10	5	4	3	2.8		2.4
(c)	What is the principle of a differential relay? What are its drawbacks and how are they overcome? Explain.							
Q3	(20 marks)						CO3	
(a)	Explain Merz Price system of protection as applied to alternator.							
(b)	Discuss a protection scheme employed for the protection of power transformers in case of an internal fault.							
(c)	Write a short note on harmonic restraint relay.							

Q4	(20 marks)	CO4,5
(a)	What are the merits and demerits of static relay over electromagnetic relay?	
(b)	Discuss Slepian's Theory. Differentiate between making and breaking capacity of circuit breaker.	
(c)	From the following data of a 50 Hz generator-emf to neutral 7.5 kV (rms), reactance of generator and connected system 4 ohm, distributed capacitance to neutral 0.01 μ F, resistance negligible: find (i) the maximum voltage across the contacts of the circuit breaker when it breaks a short-circuit current at zero current; (ii) the frequency of transient oscillation; and (iii) average rate of rise of voltage upto the first peak of oscillation.	
Q5	(20 marks)	CO5, CO6
(a)	Define Restriking voltage. Derive an expression for restriking voltage in terms of system capacitance and inductance.	
(b)	Explain, 'successive restrikes' and current chopping as applied to the interruption of capacitive and low inductive currents. How does resistance switching help in such conditions?	
(c)	A circuit breaker is rated as 2,500 A, 1,500 MVA, 33 kV, 3 s, 3-phase oil circuit breaker. Determine the rated symmetrical breaking current, rated making current, short-time rating, and rated service voltage.	

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End Semester Examination 2024

Name of the Course: B. Tech, EE

Semester: 6

Name of the Paper – Power System-II

Paper Code: TEE 602

Time: 3Hours

Maximum Marks: 100

Note: -

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question
- Total marks in each main question are twenty.
- Each question carries 10 marks

Q1	(20marks)	
(a)	What do you mean by reactance and impedance diagram? Discuss the various steps for the preparation of the reactance diagram.	
(b)	Draw reactance diagram of the power system as given below Assume base MVA and kV of the section which include G_1 are 20 MVA and 11 kV. Each generator rated voltage is 11 kV. Reactances are given in percentage.	CO1
(c)	A 10 kVA 600/300V, 50 Hz single phase transformer has primary and secondary leakage reactance each of 2.5Ω and primary and secondary resistance each of 10. Determine total impedance in p.u.	
Q2	(20 marks)	
(a)	Draw zero sequence network of the power systems as shown below.	CO2

	Prove that																													
(b)	$ I_a ^2 + I_b ^2 + I_c ^2 = 3(I_{a0} ^2 + I_{a1} ^2 + I_{a2} ^2)$ Where I_{a0}, I_{a1}, and I_{a2} are the phase sequence component of currents and I_a, I_b, and I_c are the unbalanced phasors.																													
(c)	Derive the expression for the sequence impedance of three-phase transmission line.																													
Q3	(20 marks)	CO3																												
(a)	Discuss the various types of shunt faults existing in transmission systems. Also, draw the interconnection of sequence network in LLG faults conditions.																													
(b)	The positive-, negative-, and zero sequence reactance of a 20 MVA, 13.2 kV synchronous generator are 0.3p.u, 0.2 p.u, and 0.1 p.u, respectively. The generator is solidly grounded and is not loaded. A line to ground fault occurs on phase a. Neglecting all the resistance, determine the fault current.																													
	Draw the sequence network of the power system as shown below:																													
(c)	The per unit reactances all referred to the same base are as follows:																													
	<table><tr><td></td><td>X_0</td><td>X_1</td><td>X_2</td></tr><tr><td>Generator G_1</td><td>0.05 ✓</td><td>0.3</td><td>0.2</td></tr><tr><td>Generator G_2</td><td>0.03</td><td>0.25</td><td>0.15</td></tr><tr><td>Line 1</td><td>0.70</td><td>0.3</td><td>0.3</td></tr><tr><td>Line 2</td><td>0.70</td><td>0.3</td><td>0.3</td></tr><tr><td>Transformer T_1</td><td>0.12</td><td>0.12 ✓</td><td>0.12</td></tr><tr><td>Transformer T_2</td><td>0.10</td><td>0.1</td><td>0.1</td></tr></table>		X_0	X_1	X_2	Generator G_1	0.05 ✓	0.3	0.2	Generator G_2	0.03	0.25	0.15	Line 1	0.70	0.3	0.3	Line 2	0.70	0.3	0.3	Transformer T_1	0.12	0.12 ✓	0.12	Transformer T_2	0.10	0.1	0.1	
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Q4	(20 marks)	CO4																												
(a)	Derive static load flow equations and discuss the desirable feature of an ideal load flow analysis																													
(b)	Determine Y_{bus} for the 5-bus system as shown in figure. The line series impedances between the busses are indicated in table. 																													

	Line (bus to bus) Impedance (pu) 1-2 0.25 + j 1.0 1-3 0.20 + j 0.8 1-4 0.30 + j 1.2 2-3 0.20 + j 0.8 3-4 0.15 + j 0.6	
(c)	Derive static load flow equations and discuss the desirable feature of an ideal load flow analysis	
Q5	(20 marks)	CO5, CO6
(a)	Discuss the various types of stability existing in the power systems. Explain in detail the equal area criteria of stability. Also, discuss the criterion to assess the steady state stability	
(b)	Explain in detail the equal area criteria for the estimation of the transient stability.	
(c)	Define Short Circuit Capacity (SCC) of bus of a network. Derive the expression of SCC in terms of base VA and Thevenin's impedance of network in p.u. Also, write the procedure for calculating SCC.	

EndSemester Examination 2024

Name of the Program: End Sem Semester: 6th

Name of the Course B.Tech in EE

Course Code: **TEE 611 (Special Electrical**

Machine)

Time: 3 Hour Maximum Marks: 100

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 20 marks

Q1	(20 marks)	
(a)	Discuss the Constructional feature of Brush less dc Motor in details	CO3,CO4
(b)	Discuss the merits and demerits of BLDC motor	
(b)	Enumerate the advantages and disadvantages of VRSM.	
Q2	(20 marks)	
(a)	Obtain the equation for open circuit emf of a synchronous reluctance motor	CO2,CO3
(b)	Enumerate the advantages and disadvantages of synchronous reluctance motor	
(c)	State the design parameters for stepper motor.	
Q3	(20 marks)	
(a)	State and explain the working of synchronous reluctance motor	CO1,CO3
(b)	Draw the phasor diagram of synchronous reluctance motor and hence deduce the relation for torque produced	
(c)	Compare between brushless dc motor with induction motor drives	
Q4	(20 marks)	
(a)	Draw the 2 types of static characteristics of stepper motor and explain	CO1.CO5
(b)	Deduce the mathematical relation for Low and High Magnetic Reluctances in Magnetic Circuits and discuss their influences'	
(c)	Write the merits and demerits of permanent magnet stepper motor.	
Q5	(20 marks)	
(a)	Draw the controller block diagram of stepper motor.	CO4,CO6

(b)	Discuss the Torque angle characteristic synchronous reluctance motor	
(c)	Explain the working of (8/6) switched reluctance motor	